

Sunanda Vasanthi Tata

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SUMMARY

Software Engineer with 3+ years designing and optimizing large-scale distributed systems serving 1.8M+ daily requests. Proven track record improving system performance, reducing operational friction, and shipping reliability improvements across production environments. Expertise in backend architecture, microservices design, and system optimization. Strong foundation in data structures, algorithms, and distributed system patterns. Recent focus on AI-assisted automation, including LLM-based systems and NLP pipelines. Led technical reviews, mentored engineers, and maintained high availability standards.

EDUCATION

California State University, Fullerton

Master of Science, Computer Science

Fullerton, CA

May 2026

EXPERIENCE

Pacific Life

May 2025 – Dec 2025

Software Engineer Intern

Newport Beach, CA

- Architected validation pipeline reducing manual QA effort by 40%; built **REST APIs** and **React** components processing 200K+ monthly portfolio records for 120+ concurrent users with high accuracy.
- Optimized **Kafka** stream processing pipeline ingesting 12K+ events/min with multi-layer caching, connection pooling, and async handlers; reduced **REST API** latency to sub-200ms, achieving 24% p99 response time improvement on **Kubernetes**.
- Engineered production **LLM**-based anomaly detection using **LangChain** agentic chains and **Scikit-learn** ensemble methods; deployed validation agent eliminating 40% of false positives while maintaining **OWASP** secure coding practices.
- Implemented comprehensive test coverage (unit/integration) for **React-TypeScript** frontend and **Spring Boot** backend; enforced **WCAG** accessibility compliance across bi-weekly **CI/CD** releases with zero regressions.

Hitachi Vantara

Mar 2022 – Jun 2024

Software Engineer

Bengaluru, India

- Architected **Java** microservices handling 1.8M+ daily **REST API** requests on **Azure** and **AWS** infrastructure; optimized peak-load latency from 650ms to 500ms via **PostgreSQL** query tuning, strategic indexing, and algorithmic improvements, maintaining 99.9% uptime.
- Engineered **NLP**-driven entity enrichment pipelines using **Spark/PySpark** to process web-scraped and partner-sourced data at scale; enhanced information retrieval and entity ranking pipelines for large-scale search entity systems serving billions of daily queries, improving ingestion throughput by 31%.
- Led design reviews and mentored junior engineers on distributed systems patterns; owned end-to-end feature delivery for 3 major services, deploying bi-weekly releases via **CI/CD** pipelines on **Azure DevOps** with **Terraform** infrastructure-as-code, achieving zero high-severity production incidents over 2+ years.
- Implemented **RBAC** and **OAuth 2.0** authentication across microservices; instrumented with **Prometheus** metrics and distributed tracing; reduced critical incidents by 35%, improved MTTR by 19%, ensuring compliance standards.

Accenture

Apr 2021 – Feb 2022

Software Developer

Hyderabad, India

- Designed and deployed **Python**-based **REST APIs** on a 12-node **Kubernetes** cluster with **PostgreSQL**; optimized query performance by 31% via index tuning and algorithm redesign, reducing P95 latency from 800ms to 550ms under peak load.
- Led technical design reviews for 4-person backend team on API patterns and system reliability; owned L3 incident response, achieving 95% SLA resolution via systematic root cause analysis and distributed systems expertise.
- Implemented comprehensive **unit** and **integration** tests achieving 87% code coverage; established testing best practices, reducing regression incidents by 28%.

PROJECTS

Ride-Matching and ETA Engine (Production) | Java, Python, Kafka, Kubernetes, PostgreSQL

- Architected real-time ride-matching engine with driver-ranking algorithms and geospatial search; designed event-driven microservices processing driver/rider location updates via **Redis Geo** indexing, achieving sub-**200ms** match latency while handling **10K+ concurrent requests**.

Adversarial Phishing Detection (NLP/LLM) | PyTorch, Hugging Face, LangChain, Python

- Fine-tuned transformer models (**RoBERTa**, **T5**) on 50K+ labeled phishing samples; engineered **LLM**-based agentic workflows for adversarial email classification, improving detection F1-score by 14% while maintaining 99.2% precision for production security applications.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL, TypeScript, JavaScript, C#, Go, HTML, CSS

Fundamentals: Data Structures, Algorithms, Large-Scale System Design, Distributed Systems, Networking, System Architecture, Information Retrieval

Backend and APIs: REST APIs, gRPC, Spring Boot, FastAPI, Microservices, Server-Side Rendering, Web Development

Frontend: React, TypeScript, Responsive UI, Frontend-Backend Integration, Accessibility

Cloud and DevOps: Kubernetes, Docker, AWS (EKS, RDS, S3), Azure, Terraform, CI/CD Pipelines, Feature Infrastructure

Databases: PostgreSQL, MySQL, Redis, Kafka, NoSQL, Query Optimization, Schema Design

AI and ML: LLMs, Agentic AI Workflows, LangChain, RAG Pipelines, Prompt Optimization, NLP, PyTorch, Hugging Face Transformers, Semantic Ranking, Scikit-learn, MLflow

Security and Observability: RBAC, OAuth 2.0, JWT, OWASP, WCAG, Code Quality, Testability, Resilience Patterns, Prometheus, Splunk, Centralized Logging

Testing and Quality: Unit Testing, Integration Testing, JUnit, Mockito, Code Reviews, Design Docs, Accessibility Testing